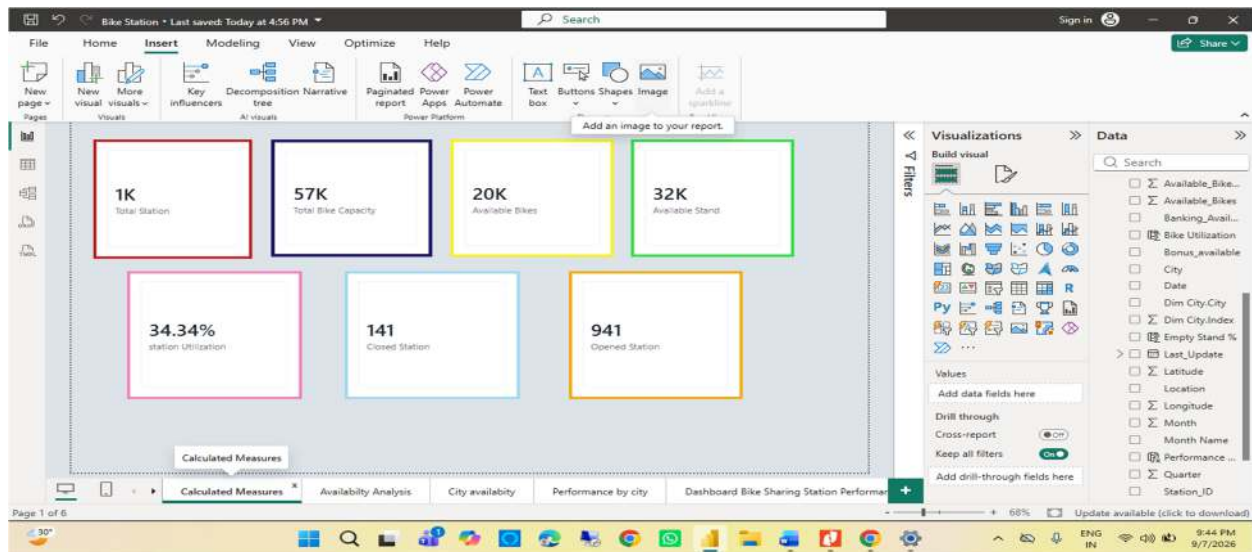


Bike Sharing Station Performance and Availability Analysis

1. Imported and explored the dataset in Power BI.
2. Cleaned and transformed data using Power Query.
3. Build a logical data model.
4. Created dimension tables and
5. Defined the relationships between the fact and dimension tables
6. Created DAX measures and calculated columns like Performance category, Bike Utilization



1 Performance Category = IF('Fact bike station'[Bike Utilization]<0.25,"Underperforming",IF('Fact bike station'[Bike Utilization]<0.60,"Average","High Performing"))

Date	Year	Month	Month Name	Quarter	Dim City.City	Dim City.Index	Bike Utilization	Empty Stand %	Performance Category
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.1	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.65	0.35	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.45	0.1	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.95	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.9	0.05	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	1	0	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.4	0.6	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0	1	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.35	0.65	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.85	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.95	0	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.9	0.1	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.95	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0	1	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.3	0.6	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.65	0.35	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.4	0.5	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.55	0.45	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.1	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.6	0.35	High Performing

Table: Fact bike station (2,856 rows) Column: Performance Category (3 distinct values)

Bike Station • Last saved: Today at 4:56 PM

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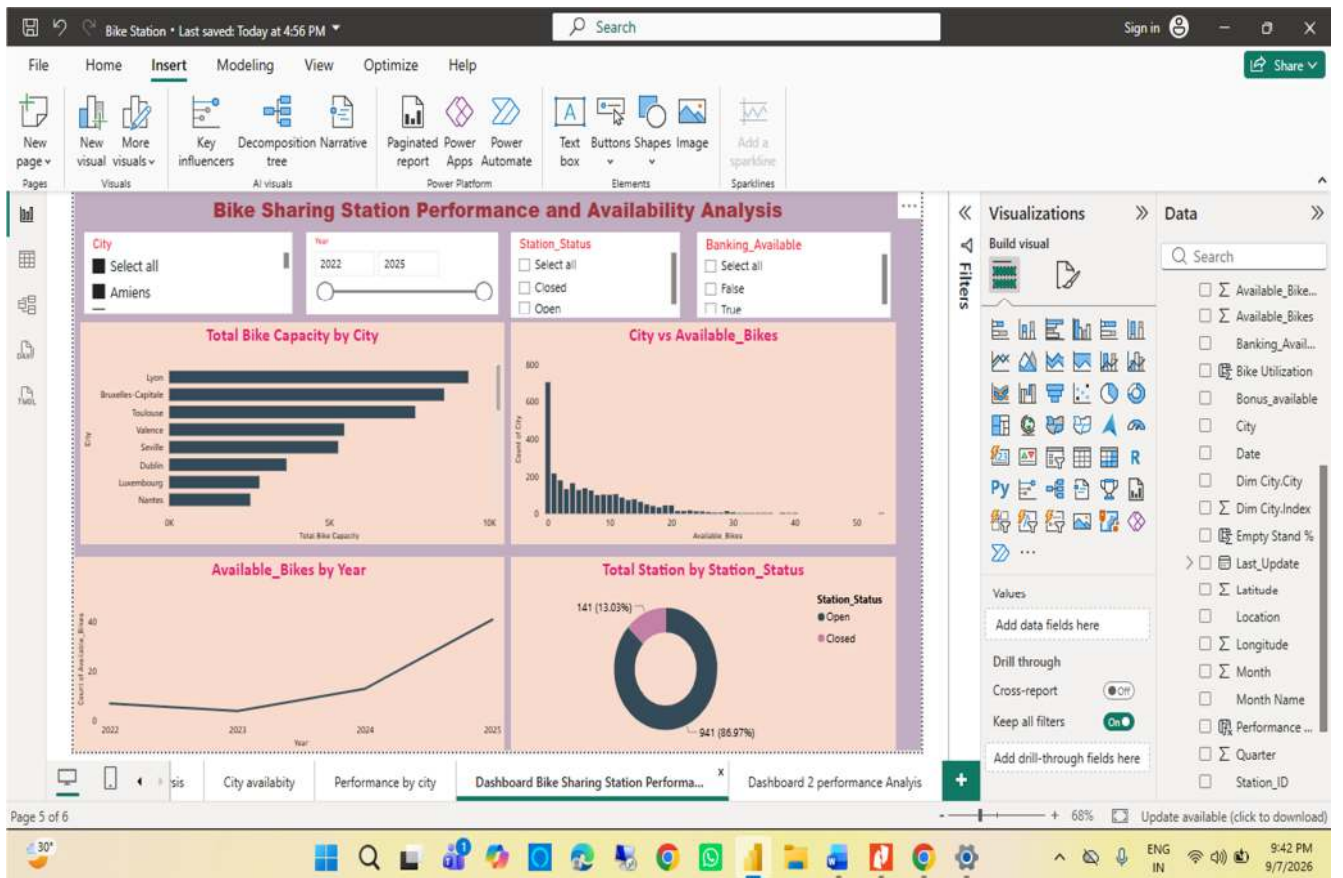
Bike Utilization = DIVIDE('Fact bike station'[Available_Bikes],'Fact bike station'[Total_Bike_Stands],0)

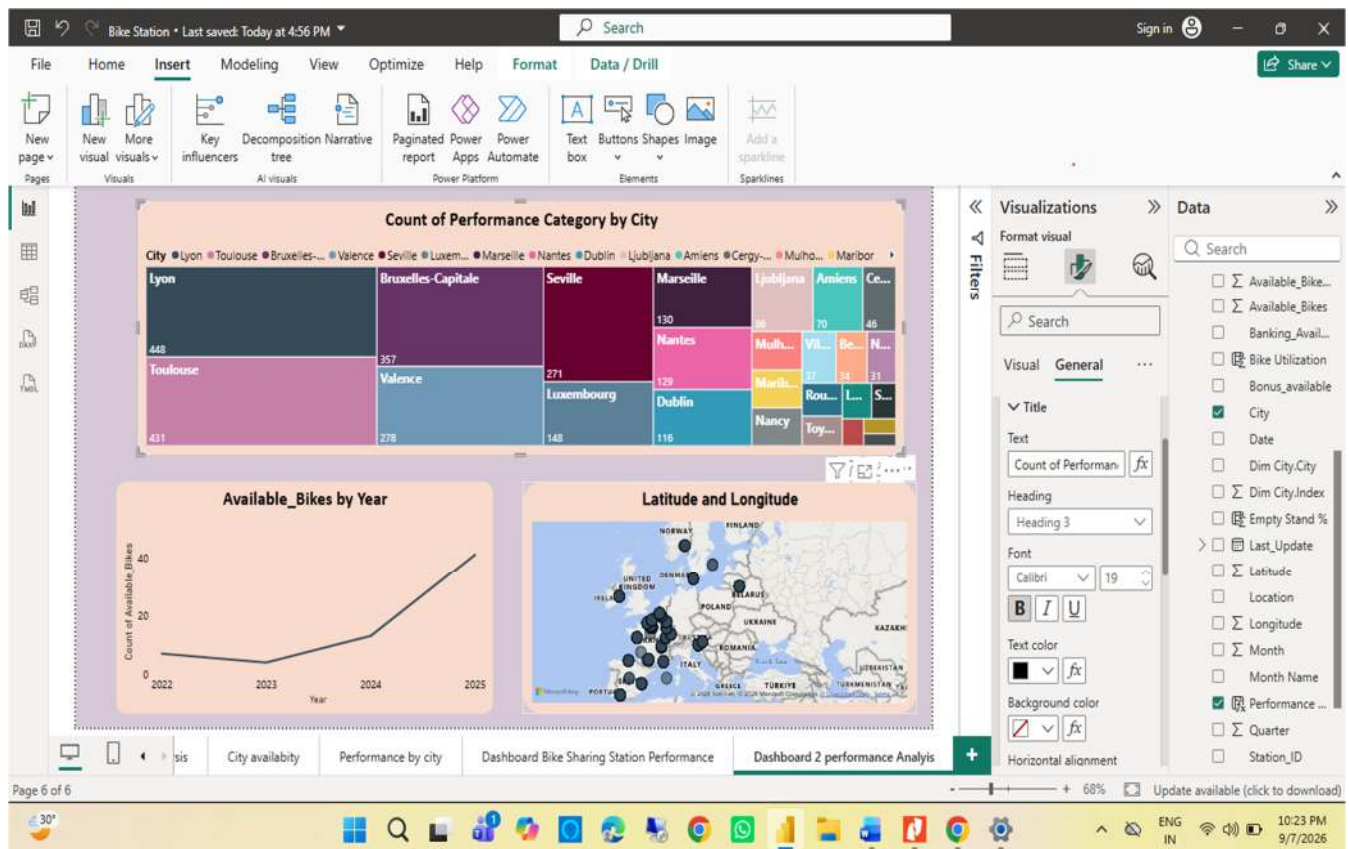
Date	Year	Month	Month Name	Quarter	Dim City.City	Dim City.Index	Bike Utilization	Empty Stand %	Performance Category
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.1	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.65	0.35	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.45	0.1	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.95	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.9	0.05	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.4	0	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0	0.6	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.35	0.65	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.85	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.95	0	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.9	0.1	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.05	0.95	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0	0.7	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.35	0.65	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.3	0.7	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.65	0.35	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.4	0.5	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.55	0.45	Average
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.1	0.9	Underperforming
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.6	0.35	High Performing
dnesday, November 12, 2025	2025	11	November	4	Seville	10	0.85	0	High Performing

Table: Fact bike station (2,856 rows) Column: Bike Utilization (290 distinct values)

Update available (click to download)

7. Designed interactive visuals and dashboards.





Key Insights:

1.Bruxelles-Capitale has the highest bike availability.

- **Available Bikes:** 3,532
- **Bike Capacity:** 8,592
- **Bike Availability %:** 41.11%
- **Insight:** Bruxelles-Capitale has the highest number of available bikes in the dataset.

2.Vilnius has the lowest bike availability.

- **Available Bikes:** 0
- **Bike Capacity:** 616
- **Bike Availability %:** 0%

3.Insight: Vilnius requires immediate operational attention because no bikes are currently available in the dataset.

Year	Available Bikes
2022	50

Year	Available Bikes
2023	11
2024	135
2025	19,450

Bike availability shows a significant increase in **2025**.

4.Lyon has the highest bike infrastructure capacity.

Vilnius, Rouen, Cergy-Pontoise, Mulhouse, Santander, and Marseille are the major underperforming cities.